

Research and Construction of a Vietnamese SMS Dataset for Machine Learning Applications.

Abstract. The scarcity of high-quality, transparently verified benchmarks severely bottlenecks the development of intelligent Vietnamese SMS spam and fraudulent detection systems. To address the issue of label noise and poor reproducibility in existing corpora, this paper introduces a quality-assured dataset construction pipeline featuring a multi-annotator validation framework and a structured disagreement resolution workflow. Within the current project phase, 2,058 raw messages were integrated from multiple sources, with a subset of 2,007 messages successfully completing the entire evaluation pipeline. Prior to formal adjudication, the independent labeling phase involving ten unique human annotators achieved an overall Fleiss' Kappa coefficient of 0.9856, signifying an almost perfect baseline agreement and validating the robustness of the proposed annotation guideline. Prior to model evaluation, an additional near-duplicate filtering stage was performed to remove redundant message templates originating from both mass spam campaigns and automated transactional notifications (e.g., banking alerts and OTP messages) that could cause train-test leakage under cross-validation. This produced an experimental subset of 1,541 unique messages for benchmark evaluation using stratified five-fold cross-validation

Keywords: Vietnamese SMS Dataset, SMS Spam Detection, Spam Filtering, Dataset Construction, Annotation Guideline, Gold Label, Inter-Annotator Agreement, Fleiss' Kappa.

1 Introduction

Short Message Service (SMS) remains an essential communication channel in Vietnam for banking, governmental services, and commercial transactions. According to DataReportal, Vietnam recorded approximately 137 million mobile cellular subscriptions in 2026, significantly exceeding the national population and highlighting the widespread adoption of mobile services [1]. However, this ubiquity has led to a persistent rise in SMS spam, creating major cybersecurity challenges. Attackers increasingly employ sophisticated social engineering techniques, impersonating banks and trusted organizations to deceive users into revealing sensitive data [3, 4]. Consequently, developing reliable automated detection systems has attracted substantial attention from both academia and industry.

Although significant progress has been made in Vietnamese SMS classification, the availability of high-quality benchmark datasets remains limited. Most existing public corpora, such as the SMS Spam Collection [6], were developed for English and do not represent the linguistic nuances, localized abbreviations, or specific scam patterns observed in Vietnamese. Furthermore, existing regional datasets are often limited in scale, proprietary, or lack transparent documentation regarding annotation

quality control. Many lack clear disagreement resolution procedures, which undermines dataset reproducibility, annotation consistency, and long-term research value [7].

To address these limitations, this paper presents a quality-assured, transparent framework for constructing a Vietnamese SMS dataset. Rather than relying solely on simplistic majority voting, our methodology prioritizes annotation integrity by combining independent multi-annotator labeling with a dual-stage verification process to establish high-fidelity Gold Labels. Messages reaching a consensus among ten independent annotators are directly integrated, while disagreement cases undergo collaborative review and root-cause analysis. Currently, the framework has processed 2,058 messages, successfully validating and finalizing a baseline subset of 2,007 high-quality Vietnamese SMS records for downstream machine learning research.

2 Related work

While substantial efforts have been made to construct benchmark corpora, existing datasets exhibit significant limitations regarding annotation transparency and quality control. For instance, the widely used SMS Spam Collection v.1 for English (Almeida et al.) and the prior Vietnamese dataset proposed by Pham and Le-Hong (2016) rely on manual aggregation but fail to report formal Inter-Annotator Agreement (IAA) metrics or structured disagreement resolution protocols. This lack of transparent quality assurance undermines dataset reproducibility and the reliability of the ground-truth labels. Furthermore, existing works often overlook the critical issue of near-duplicate template-based campaigns, which can lead to train-test leakage and artificially inflated model performance. In contrast, our proposed dataset directly addresses these gaps by establishing a rigorous multi-annotator framework (achieving a Fleiss' Kappa of 0.9856) and a strict deduplication pipeline prior to benchmark evaluation

Dataset	Language	Size (Records)	Taxonomy	Inter-Annotator Agreement	Public Availability
SMS Spam Collection v.1 (Almeida et al.)	English	5574	Ham/Spam	Not Reported	Yes
Pham & Le-Hong (2016)	Vietnamese	6599	Ham/Spam	Not reported	Upon request
Ours work	Vietnamese	2007	Ham/Spam	Fleiss' Kappa = 0.9856	Yes

Table 1. Comparison of the proposed dataset with existing SMS spam corpora.

3 Proposed Dataset Construction Framework

3.1 Overall Framework

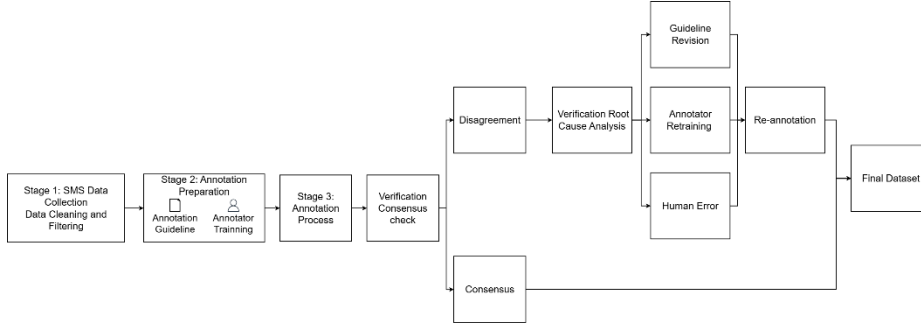


Fig. 1. Overall workflow of the proposed quality-assured Vietnamese SMS dataset construction framework.

The primary objective of this study is not only to construct a Vietnamese SMS dataset but also to establish a quality-assured and reproducible annotation framework for generating reliable ground-truth labels. Instead of treating annotation as a one-time labeling activity, the proposed framework incorporates continuous quality improvement throughout the dataset construction process. The overall workflow is illustrated in Fig. 1.

The proposed framework consists of four major stages: SMS data collection, annotation preparation, independent multi-annotator labeling, and iterative quality assurance. SMS messages are first collected from multiple heterogeneous sources and filtered according to predefined inclusion and exclusion criteria. Prior to annotation, all annotators are provided with a standardized Annotation Guideline to ensure a common understanding of annotation objectives and decision rules.

Inspired by the systematic disagreement resolution procedure proposed by Oortwijn et al. [7], the proposed framework treats annotation disagreements as indicators of potential quality issues rather than simple annotation errors. Each disagreement is therefore analyzed to identify its underlying cause, including ambiguous annotation guidelines, conceptual misalignment among annotators, inconsistent annotation behavior, or individual annotation mistakes. Appropriate corrective actions, such as guideline refinement, annotator retraining, or re-annotation, are subsequently performed before the messages are re-evaluated through the verification process.

Unlike conventional annotation pipelines, the proposed framework establishes a closed-loop quality assurance mechanism, in which disagreement analysis continuously improves both the annotation guideline and annotator consistency. This iterative process aims to maximize annotation reliability while ensuring that the resulting da-

taset remains transparent, reproducible, and suitable for future machine learning and natural language processing research.

3.2 SMS Collection

The quality of a machine learning dataset largely depends on the diversity, authenticity, and representativeness of its data sources. Therefore, the proposed dataset was constructed by collecting Vietnamese SMS messages from multiple heterogeneous sources rather than relying on a single collection channel. This strategy aims to improve linguistic diversity while reducing potential sampling bias.

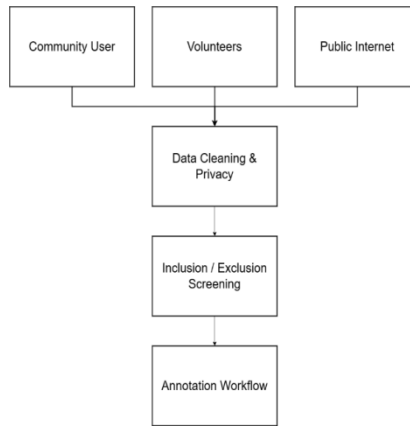


Fig. 2. SMS Data Collection Pipeline

As illustrated in Fig. 2, SMS messages were collected from four primary channels: community contributions, volunteer submissions, publicly available online resources, and partner organizations. In the dataset schema (Table 4), these sources are mapped into three standardized categorical attributes: *crawl* (online resources), *user_report* (community and volunteer submissions), and *whitelist* (partner organizations). Community contributions and volunteer submissions provide real-world messages received by mobile users in their daily lives, allowing the dataset to capture contemporary phishing campaigns, promotional advertisements, financial notifications, and legitimate personal communications.

To protect user privacy, all collected messages underwent an anonymization process before annotation. Personally identifiable information (PII), including phone numbers, personal names, identification numbers, bank account information, URLs, and othersensitive content, was either removed or replaced with standardized placeholders while preserving the semantic characteristics required for annotation and subsequent machine learning applications.

To ensure high data quality and eliminate potential noise before the formal annotation phase, a rigorous screening pipeline was executed combining automated scripts and a systematic manual inspection protocol. Out of the 2,058 raw messages initially-

collected from the four heterogeneous sources, a total of 51 records (2.48%) were systematically excluded based strictly on the predefined technical and semantic filtering criteria outlined in Table 2. First, an automated Python script was utilized to perform exact deduplication, removing 23 redundant message variants generated by mass-broadcast spam campaigns. Second, a structured, two-pass manual verification was conducted by the core research team to eliminate 9 messages with severe structural corruption (e.g., truncated text or unresolvable encoding failures) and 19 messages holding insufficient semantic information following the PII anonymization phase. This hybrid screening protocol ensures that the subsequent multi-annotator evaluation pipeline operates exclusively on a clean, robust, and linguistically viable corpus, reinforcing the reproducibility of our methodology.

Reason	Messages
Duplicate	23
Severe structural corruption, including truncated text from system logging errors or unresolvable character encoding failures	9
Contained insufficient semantic information after the anonymization process (e.g., messages stripped of critical non-textual tokens or containing only isolated, meaningless numeric OTP codes).	19

Table 2. Quantitative breakdown of excluded messages during the data cleaning and filtering stage.

3.3 Annotation Guideline

To minimize annotation subjectivity and improve labeling consistency, a standardized Annotation Guideline was developed before the annotation process. The guideline establishes a unified decision-making framework by defining annotation objectives, label definitions, decision rules, and representative examples. All annotators were required to study the guideline before participating in the annotation task.

The guideline also addresses key linguistic nuances in Vietnamese SMS communication. As observed by Tuan et al. [4], spammers overwhelmingly favor unaccented text (non-diacritic Vietnamese) to bypass character limit constraints and reduce transmission costs while preserving readability for human targets. Our guideline incorporates standard normalization rules (e.g., mapping variant promotional markers such as 'quảng cáo', 'quang cao', '[QC]' to a unified token representation [4]) and specifies decision rules for ambiguous cases. Annotators were instructed to evaluate each message according to its communication intent, contextual meaning, and potential security risk rather than isolated keywords. For example, the presence of terms such as promotion, bank, or reward does not automatically indicate spam. Instead, annotators considered whether the overall message exhibited deception, impersonation, coercion, or unauthorized commercial solicitation.

Spam is defined as a broad category comprising two semantic sub-categories: Commercial Spam, which includes unsolicited marketing promotions and unauthorized

advertisements, and Fraudulent/Phishing Spam, which includes impersonation attacks, fraudulent financial schemes, and malicious messages intended to steal information or financial assets.

For the baseline binary classification task, both sub-categories are assigned the same Spam label. This design reflects the objective of constructing a binary SMS spam detection dataset while preserving finer semantic distinctions in the annotation guideline for future analyses and potential taxonomy extension. The guideline also specifies decision rules for ambiguous cases. Annotators were instructed to evaluate each message according to its communication intent, contextual meaning, and potential security risk rather than isolated keywords. For example, the presence of terms such as promotion, bank, or reward does not automatically indicate spam. Instead, annotators considered whether the overall message exhibited deception, impersonation, coercion, or unauthorized commercial solicitation.

Representative examples and common annotation pitfalls identified during preliminary annotation rounds were incorporated into the guideline to improve inter-annotator consistency. During the verification stage, the guideline was revised whenever systematic ambiguities were identified through disagreement analysis, supporting continuous improvement of annotation quality.

3.4 Independent Multi-Annotator Labeling

After completing the annotation preparation stage, the collected SMS messages were independently annotated by a team of trained annotators following the standardized Annotation Guideline described in the previous section. Independent annotation was adopted to minimize potential bias arising from communication among annotators and to ensure that each labeling decision reflected the annotator's own interpretation of the predefined annotation criteria.

Each SMS message was assigned to ten independent annotators, all of whom evaluated the message without access to the labels assigned by other annotators. This independent annotation strategy provides multiple judgments for each message, allowing the subsequent verification process to evaluate annotation consistency and identify disagreement cases objectively. The annotation team comprised 10 native Vietnamese speakers (undergraduate and graduate researchers in Computer Science and Cybersecurity). All annotators completed a formal training session on the Annotation Guideline prior to labeling. Annotators were compensated fairly for their effort and signed privacy non-disclosure agreements to protect user data.

Unlike majority-voting approaches that determine the final label immediately after annotation, the proposed framework intentionally separates annotation from verification. The objective of this design is to preserve all individual annotation decisions for subsequent quality assessment rather than discarding minority opinions. Consequently, every annotation record, including disagreement cases, becomes valuable evidence for evaluating annotation consistency and identifying potential weaknesses in the annotation process.

After all annotators completed their labeling tasks, the annotation results were merged into a unified annotation repository. At this stage, no annotation was manual-

ly modified or removed. Instead, the merged annotation results served as the input for the dual-stage verification framework presented in the following section.

3.5 Consensus Assessment and Disagreement Resolution

Following independent annotation, results are subjected to a structured consensus assessment to evaluate consistency. Unlike conventional workflows that rely on majority voting, the proposed framework preserves every individual decision and systematically examines disagreement cases to identify quality issues.

Messages achieving complete consensus are directly accepted into the final dataset. Conversely, disagreement cases are transferred to a dedicated analysis process. Inspired by Oortwijn et al. [7], this analysis evaluates multiple perspectives, including guideline ambiguity, conceptual inconsistency, variable annotation behavior, and human error, to address root causes systematically rather than case-by-case.

3.6 Ethics Statement and Data Licensing

Ethics Statement: All SMS messages were collected with user consent or obtained from public reporting platforms. Sensitive personal details—including phone numbers, personal names, account credentials, and specific URLs—were systematically replaced with standardized placeholders (e.g., [PHONE], [URL], [NAME]) during anonymization to safeguard user privacy.

Licensing: The finalized dataset and annotation guidelines are published under the Creative Commons Attribution 4.0 International (CC BY 4.0) license. The benchmark baseline code and utility scripts are distributed under the MIT License.

3.7 Dataset Structure and Format

To ensure absolute transparency, data portability, and high reproducibility for the research community, the finalized Vietnamese SMS phishing dataset is structured into a standardized hierarchy. The repository package is organized as follows:

Component name	Type	Description / Purpose
anonymous-sms-dataset/	Root directory	The primary folder encapsulating all published assets, code dependencies, and data resources.
anonymous-sms-dataset/README.md	File	Provides instructions for installation, usage examples, and licensing information.
anonymous-sms-dataset/annotation_guidelines.pdf	File	The strict protocol defining syntactic and semantic rules applied by the ten independent annotators.

data/	Sub-Folder	Dedicated container holding all operational data partitions and computational metadata.
data/raw_corpus.csv	File	The initial unadjudicated corpus comprising 2,058 collected records across all integrated sources.
data/gold_standard.csv	File	The final benchmark dataset containing 1,763 fully adjudicated messages used for downstream evaluation.
data/metadata_distribution.json	File	Detailed voting matrix preserving individual label decisions from the ten human annotators.
Data/benchmark_subset.csv	File	A leakage-controlled evaluation subset containing 1,541 SMS messages after near-duplicate removal, used for baseline model benchmarking.

Table 3. Directory structure, schema specifications, and components of the dataset repository.

The core utility of the repository resides within the `gold_standard.csv` partition. To provide a clear overview of the data architecture for machine learning integration, Table 3 outlines the explicit attribute schema, data types, constraints, and academic examples for each feature included in the gold-standard benchmark file.

File name	Attribute name	Data type	Description	Example
raw_corpus.csv	message_id	Integer	Unique identifier for tracking.	COM_001
	raw_text	String	The original raw SMS text.	<i>“quy khach co hoa don tien dien can thanh toan...”</i>
	source	String	Origin repository (crawl, user_report, whitelist).	<i>“internet”, “user_report”,...</i>
	gold_label	String	The definitive adjudicated class	HAM or SPAM
	agreement	Integer	Matching	<i>“10”, “9”, “8”,...</i>

			votes among the 10 annotators.	
gold_standard.csv	text	String	Streamlined text content ready for tokenization.	"Chuc mung TB 093xxx trung thuong..."
	label	String	Clean target label for model training	HAM or SPAM
benchmark_subset.csv	text	String	Preprocessed SMS text selected for leakage controlled benchmark evaluation after near-duplicate removal.	"Chuc mung TB 093xxx trung thuong..."
	label	String	Clean target label for model training	Ham or Spam

Table 4. Attribute schema specifications across dataset repository partitions (raw_corpus.csv, gold_standard.csv, and benchmark_subset.csv).

4 Experimental Results

4.1 Initial Annotation Results

Following the independent annotation process described in Section 2, ten annotators completed the labeling task using the standardized Annotation Guideline. At this stage, the annotation results represent the initial labeling outcomes before the disagreement resolution and quality assurance procedures were applied. Therefore, the statistics presented in this section are intended to evaluate the consistency of the annotation process rather than the characteristics of the final released dataset.

SMS Annotated	2007
Annotator	10
Label per message	10
Total annotation decisions	20070

Table 5. Initial Annotation Summary

4.2 Inter-Annotator Agreement

Following the first-stage annotation process, each SMS message was independently labeled by ten annotators according to the Annotation Guideline described in Section 2. To evaluate annotation consistency, the individual annotation files were merged by SMS identifier, allowing the labels assigned by different annotators to be aligned and compared on a message-by-message basis.

The merged annotation matrix was evaluated using Fleiss' Kappa, a statistical measure designed to assess agreement among multiple annotators assigning categorical labels to the same set of instances [8, 10]. Unlike Cohen's Kappa, which is mathematically restricted to a fixed pair of exactly two raters [5, 9], Cohen's framework is fundamentally unsuitable for this study because each SMS message was independently annotated by a larger group of ten annotators. Consequently, Fleiss' Kappa is uniquely appropriate as it accommodates an arbitrary number of raters per item and estimates the degree to which the observed consensus exceeds the chance-expected agreement [8].

The Fleiss' Kappa coefficient is computed as follows:

$$\kappa = \frac{\bar{P} - \bar{P}_e}{1 - \bar{P}_e} \quad (1)$$

Where $\bar{P}$ denotes the observed agreement among annotators and $\bar{P}_e$ represents the expected agreement by chance. The observed agreement is obtained by averaging the agreement score of each SMS message, whereas the expected agreement is estimated from the overall distribution of annotation categories

Metric	Value
Total annotated SMS	2007
Number of annotators	10
Annotation categories	2
Messages with complete agreement	1963
Messages with disagreement	44
Fleiss' Kappa	0.9856
Agreement level (Landis & Koch)	Almost perfect

Table 6. Summary of inter-annotator agreement before disagreement resolution.

As shown in Table 6, the proposed annotation framework achieved a Fleiss' Kappa value of 0.9856, corresponding to Almost Perfect Agreement according to the interpretation proposed by Landis and Koch [5]. This result indicates that the Annotation Guideline enabled a high degree of consistency among independent annotators prior to the disagreement resolution process. Nevertheless, Fleiss' Kappa provides only an overall measure of annotation consistency and does not reveal how agreement is distributed across individual SMS messages.

To provide a more intuitive view of the annotation outcomes, Fig. 3 illustrates the distribution of agreement levels among the 2007 annotated SMS messages before disagreement resolution.

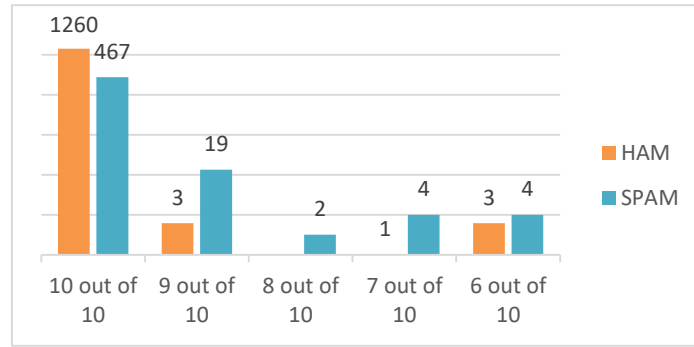


Fig. 3. Distribution of inter-annotator agreement before disagreement resolution.

Agreement Level	Ham	Spam	Total
10/10	1,259	704	1,963 (97.81%)
9/10	3	20	23 (1.15%)
8/10	2	4	6 (0.30%)
7/10	3	7	10 (0.50%)
6/10	0	5	5 (0.25%)
Total	1,267	740	2,007 (100.00%)

Table 7. Detailed distribution of inter-annotator agreement levels across classes prior to disagreement resolution.

4.3 Disagreement Analysis

Although the proposed annotation framework achieved an exceptionally high baseline inter-annotator agreement (Fleiss' Kappa = 0.9856, validating the initial Guideline and training efficacy), a minor subset of 44 SMS messages (2.19%) produced inconsistent labels among the ten independent annotators prior to resolution. Rather than treating these contested cases as statistical noise or discarding minority votes, they were transferred to a dedicated disagreement queue to analyze root causes and generate the final Gold Standard.

Table 8 summarizes the distribution of root causes identified during the systematic review process, along with the specific corrective actions executed.

Disagreement source	Cases	Percentage
Guideline ambiguity	28	63,64%
Conceptual misunderstanding	11	25,00%
Individual oversight	5	11,36%

Table 8. Root-cause classification and resolution strategies for the 44 contested messages

To demonstrate the concrete utility of our closed-loop quality assurance framework, all 44 contested instances underwent formal adjudication following guideline refinement. As detailed in Table 7, the resolution process successfully eliminated all semantic alignment anomalies. Specifically, 27 messages were reclaimed as legitimate Ham communications (primarily non-standard transaction alerts and automated system announcements that initial annotators misclassified due to strict keyword biases), while 17 messages were stabilized under the Spam class (subtle phishing variants containing obfuscated URLs or promotional teencode)

Metric	Round 1	Round 2	Relative improvement
Total Vetted Messages	2007	2007	Complete dataset retained.
Absolute Consensus (10/10)	1963	2007	+44 resolved messages (+2.19%).
Active Disagreement (<10/10)	44	0	100% semantic noise eliminated.

Table 9. Data fidelity progression: Initial Inter-Annotator Agreement (Round 1) vs. Final Gold Standard Generation (Post-Resolution).

By systematically routing borderline cases through our disagreement resolution work-flow, 100% of the collected corpus (2,007 messages) was converted into a high-fidelity Gold Standard dataset containing 1,286 Ham (1,259 + 27) and 721 Spam (704 + 17) records. This "before vs. after" progression empirically proves that our framework does not merely log human divergence, but actively cures label noise to deliver clean training signals for downstream machine learning applications

4.4 Baseline Classification Results

4.4.1. Experimental Setup and Implementation Details

To establish a transparent, highly reproducible performance benchmark for Vietnamese SMS spam detection, we conducted a comparative empirical study using the finalized Gold Standard corpus. Before model training, a dedicated near-duplicate filtering stage was applied to remove template-based SMS messages that could introduce train-test leakage. The resulting evaluation corpus contained 1,541 unique SMS messages. The classification workflow evaluates two distinct families of algorithms: traditional machine learning architectures utilizing surface-level features and a fine-tuned deep contextual Transformer model [11].

For traditional machine learning baselines, including Support Vector Machine with Linear and RBF kernels [12], Logistic Regression [13], Multinomial Naive Bayes [14], and a Multi-Layer Perceptron [15], textual representations were extracted using Term Frequency-Inverse Document Frequency (TF-IDF) vectorization implemented via scikit-learn [16]. The feature space was configured with an n-gram range of (1, 2) and capped at a maximum vocabulary size of 5,000 features. Prior to feature extraction, Vietnamese compound words were segmented using the PyVi toolkit (ViTokenizer.tokenize) [17]. However, because PyVi relies on standard dictionary lookup,

its segmentation efficacy degrades on unaccented SMS text and informal teencode (e.g., 'ck' for 'chuyển khoản', 'ngkhac' for 'người khác'), introducing tokenization noise into traditional surface feature representations.

To fulfill our promise of evaluating deep pre-trained language models, we integrated vinai/phobert-base (PhoBERT-base), a state-of-the-art pre-trained Transformer architecture optimized for Vietnamese natural language processing. Fine-tuning of PhoBERT-base was executed on a dedicated workstation equipped with an NVIDIA GeForce RTX 3060 GPU running Python 3.14.2 and PyTorch 2.12.0+cu126. The model was trained for 4 epochs using the AdamW optimizer with a learning rate of 0.00002 (2e-5), a weight decay of 0.01, a warmup ratio of 0.1, a per-device batch size of 16, and a maximum sequence length of 128 tokens. Model selection was governed by evaluating the highest F1-score on the positive class at each epoch (load_best_model_at_end=True).

4.4.2. Near-Duplicate Filtering and Train-Test Leakage Protection

While the deduplication process was initially motivated by the need to filter out template-based phishing and spam campaigns, empirical observations revealed that the vast majority of near-duplicates (87%) were actually legitimate messages (Ham). This is highly consistent with real-world mobile traffic, where legitimate institutional communications, such as banking OTPs, automated transaction alerts, and promotional broadcasts, rely heavily on fixed templates.

To eliminate potential evaluation bias caused by near-duplicate spam campaigns, a rigorous multi-stage deduplication pipeline was executed on the 2,007 gold-standard messages prior to any dataset splitting. Phishing campaigns typically generate vast amounts of template-based messages that differ only in dynamic entities (e.g., recipient names, phone numbers, or fraudulent URLs). Performing cross-validation directly on raw data would risk placing near-identical variants into both training and testing folds, leading to artificially inflated performance metrics. To prevent this, we employed a character-level 3-gram Jaccard similarity metric. Texts were first normalized (lowercased and whitespace-trimmed) and decomposed into sets of contiguous 3-character sequences. The Jaccard index between two message sets S_A and S_B was computed as:

$$J(S_A, S_B) = \frac{|S_A \cap S_B|}{|S_A \cup S_B|} \quad (2)$$

Messages exhibiting a similarity threshold of $J \geq 0.85$ were identified as campaign variants and filtered out. This stage successfully removed 466 near-duplicate records, yielding a strictly unique benchmark corpus of 1,541 messages (878 Ham and 663 Spam). This workflow guarantees that no message pair with $J \geq 0.85$ spans across the training and evaluation splits of any fold.

4.4.3 Comparative Performance Analysis

Table 10 consolidates the predictive performance metrics evaluated on the independentevaluated using stratified five-fold cross-validation. All reported metrics,

including Accuracy, Precision, Recall, and F1-score, were computed for the Spam class for the Spam positive class (Class 1) using the harmonic mean formula:

$$F_1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (3)$$

Model Architecture	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
PhoBERT-base	97.99 ±0.91	97.74 ±1.25	97.59 ±1.00	97.66 ±1.05
SVM (Linear)	96.69 ±1.86	95.21 ±3.12	97.29 ±1.31	96.22 ±2.10
SVM (RBF)	96.88 ±1.57	94.97 ±2.98	98.04 ± 0.60	96.46 ±1.75
Multi-Layer Perceptron	96.30 ±1.69	94.62 ±2.73	96.98 ± 1.36	95.77 ±1.90
Logistic Regression	96.30 ±1.50	94.87 ±2.54	96.68 ± 1.02	95.76 ±1.68
Multinomial Naive Bayes	93.58 ±1.02	94.25 ±1.62	90.65 ± 2.71	92.38 ±1.28

Table 10. Baseline classification performance comparison on the unique dataset (1,541 records) evaluated using stratified five-fold cross-validation (Mean ± Std).

As detailed in Table 10, the fine-tuned deep contextual Transformer model, PhoBERT-base, achieved the highest overall classification efficacy across the 5 folds, attaining a mean cross-validation accuracy of $97.99\% \pm 0.91\%$, a precision of $97.74\% \pm 1.25\%$, a recall of $97.59\% \pm 1.00\%$, and an F1-Score of $97.66\% \pm 1.05\%$. To further evaluate statistical robustness independently of fold partitioning, a non-parametric Bootstrap resampling ($B = 1000$) was conducted on out-of-fold predictions. As reported in Table 11, PhoBERT-base consistently maintained the highest lower-bound performance, yielding a narrow 95% F1-score confidence interval of [96.85%, 98.42%].

Model Architecture	Accuracy 95% CI (%)	Precision 95% CI (%)	Recall 95% CI (%)	F1-Score 95% CI (%)
PhoBERT-base	[97.27%, 98.64%]	[96.59%, 98.79%]	[96.38%, 98.64%]	[96.85%, 98.42%]
SVM (Linear)	[95.78%, 97.47%]	[93.52%, 96.55%]	[96.02%, 98.44%]	[95.15%, 97.10%]
SVM (RBF)	[95.98%, 97.73%]	[93.26%, 96.45%]	[96.97%, 99.07%]	[95.36%, 97.38%]
Multi-Layer Perceptron	[95.33%, 97.14%]	[92.93%, 96.11%]	[95.59%, 98.17%]	[94.63%, 96.76%]
Logistic Regression	[95.33%, 97.14%]	[92.95%, 96.40%]	[95.32%, 97.91%]	[94.62%, 96.75%]
Multinomial	[92.34%,	[92.24%,	[88.22%,	[90.73%,

Naive Bayes	94.81%]	95.94%]	92.73%]	93.83%]
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Table 11. Bootstrap 95% Confidence Intervals (B=1000 resamples) computed on out-of-fold predictions.

Bootstrap resampling was performed on the aggregated out-of-fold predictions generated during five-fold cross-validation. Each bootstrap sample was obtained by sampling prediction pairs with replacement 1,000 times. Among traditional machine learning baselines, SVM with RBF kernel achieved the highest overall recall ($98.04\% \pm 0.60\%$) with an F1-Score of $96.46\% \pm 1.75\%$, closely followed by Linear SVM ($96.22\% \pm 2.10\%$). Logistic Regression and MLP demonstrated stable convergence around $\sim 95.76\%$ F1-score. Conversely, Multinomial Naive Bayes registered the lowest performance ($92.38\% \pm 1.28\%$ F1-score) due to a noticeable degradation in recall ($90.65\% \pm 2.71\%$). This performance drop stems from two primary factors: (i) the strong conditional independence assumption failing to model complex word dependencies in multi-keyword scam messages, and (ii) static TF-IDF feature spaces failing to generalize across misspelled teencode variations. These empirical findings corroborate the conclusions of Tuan et al. [4], who similarly demonstrated that pre-trained contextual embeddings (PhoBERT) significantly outperform traditional surface-level feature models (such as Naive Bayes and SVM) across diverse Vietnamese text variants [4]. In our work, PhoBERT-base leverages subword Byte-Pair Encoding (BPE) tokenization and deep contextual attention, enabling it to map noisy teencode variants to their underlying semantic representations effectively within a strictly deduplicated, leakage-controlled evaluation framework.

4.5 Limitations

While the proposed framework establishes a transparent, quality-assured dataset for Vietnamese SMS classification, several research limitations should be acknowledged:

1. In-Distribution Evaluation Scope: Model performance metrics evaluate five-fold cross-validation on an in-distribution, deduplicated benchmark corpus. Real-world performance on out-of-distribution (OOD) traffic or newly emerging zero-day phishing campaigns employing novel lexical obfuscation tactics may vary.
2. Dataset Scale and Longitudinal Expansion: The dataset currently encapsulates 2,007 gold-standard messages (1,541 unique records). Although sufficient for establishing robust baseline classifiers, continuous longitudinal collection is necessary to track evolving mobile threats over time.
3. Morphological Segmentation Bottlenecks: Standard Vietnamese segmenters (e.g., PyVi) exhibit reduced accuracy on non-diacritic SMS text and informal teencode, emphasizing the necessity of domain-adapted subword language models.

5 Conclusion and future work

This study establishes a quality-assured and transparent annotation methodology for constructing a Vietnamese SMS phishing dataset, shifting the emphasis from simple data accumulation to reproducible quality control. Rather than dismissing human disagreement as statistical noise, the proposed framework leverages annotator divergence as a critical linguistic signal to uncover semantic ambiguities and systematically refine the underlying gold standard labels. Prior to formal adjudication, the independent labeling phase involving ten unique annotators achieved an overall Fleiss' Kappa coefficient of 0.9856, signifying an 'Almost Perfect Agreement.' This high baseline consensus confirms that the standardized annotation guidelines effectively established clear decision boundaries for the vast majority of messages. Human disagreement was strictly confined to a minor 2.19% subset of edge cases, primarily caused by non-standard banking notification templates and localized teencode abbreviations. By routing these specific borderline instances through the structured disagreement resolution workflow, the methodology effectively guarantees the integrity of the finalized dataset.

At the current developmental stage, the corpus comprises 2,058 collected messages, of which 2,007 have successfully progressed through the complete multi-annotator validation pipeline (yielding 1,541 unique records post-deduplication). To ensure absolute transparency and facilitate open-science collaboration, the final benchmark dataset is packaged alongside its standardized Annotation Guideline, granular disagreement metadata, and open-access distribution formats. Future initiatives will actively expand this corpus beyond its current phase to achieve a targeted milestone of over 5,000 annotated Vietnamese SMS messages, specifically focusing on capturing newly emerging zero-day phishing variants to stress-test and evolve the resolution framework continuously. Ultimately, this scaling corpus serves as a foundational, high-fidelity benchmark for training and evaluating next-generation intelligent cybersecurity detection systems within the Vietnamese natural language processing domain.

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